Solution:

The required network for the above information.

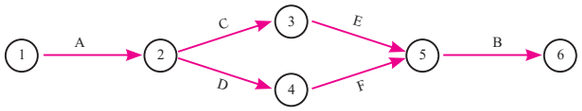


Fig 10.5

Example 10.10

Develop a network based on the following information:

Activity:	A	B	C	D	E	F	G	H
Immediate predecessor:	-	-	A	B	C,D	C,D	E	F

Solution:

Using the immediate precedence relationships and following the rules of network construction, the required network is shown in following figure.

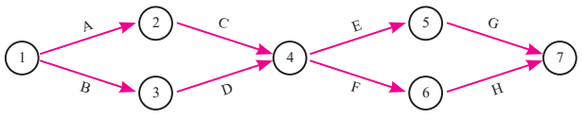


Fig 10.6

Dummy activity:

An activity which does not consume any resource or time, but merely depict the technological dependence is called a dummy activity. It is represented by dotted lines.

Example 10.11

Draw a network diagram for the project whose activities and their predecessor relationships are given below:

Activity :	A	B	C	D	E	F	G	H	I	J	K
Predecessor activity:	-	-	-	A	B	B	C	D	F	H,I	F,G

Solution:

Using the precedence relationships and following the rules of network construction, the required network diagram is shown in following figure

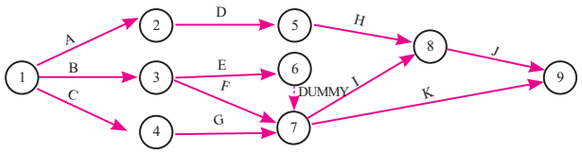


Fig 10.7

Example 10.12

Construct a network diagram for the following situation:

$A < D, E;$ $B, D < F;$ $C < G$ and $B < H.$

Solution:

Using the precedence relationships and following the rules of network construction, the required network is shown in following figure.

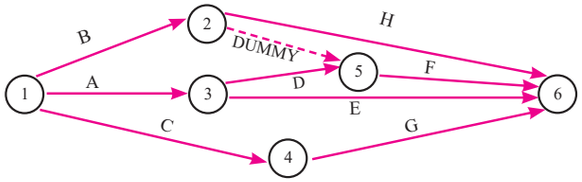


Fig 10.8

10.2.2 Critical path analysis

For each activity an estimate must be made of time that will be spent in the actual accomplishment of that activity. Estimates may be expressed in hours, days, weeks or any other convenient unit of time. The time estimate is usually written in the network immediately above the arrow. For the purpose of calculating various times of events and activities, the following terms shall be used in critical path calculations:

E_i = Earliest start time of event i

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